

UQFF SNR Buoyancy Master Equation — Five Supernova Remnant Systems

Daniel T. Murphy

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PAPER_485: UQFF SNR Buoyancy Master Equation — Five Supernova Remnant Systems

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Abstract

This paper presents the Unified Quantum Force Field (UQFF) buoyancy master equation for five supernova remnant (SNR) and high-energy systems: SN 1006 (Type Ia remnant), Eta Carinae (luminous blue variable), Chandra Archive Collection, Galactic Center (near Sgr A*), and Kepler's SNR (SN 1604). The master buoyancy force $F_{U,Bi,i}$ is decomposed into six physically distinct components: LENR (low-energy nuclear reaction), activation energy, dark energy pressure, resonance, neutron production, and relativistic correction. A dual-mode quadratic solver provides SNR expansion mode separation.

1. Master Buoyancy Equation

$$F_{U,Bi,i} = F_{LENR} + F_{act} + F_{DE} + F_{res} + F_{neutron} + F_{rel}$$

1.1 Component Equations

F_{LENR} (Low-Energy Nuclear Reaction Force):

$$F_{LENR} = k_{LENR} \cdot \rho_{vac} \cdot \sin(\omega_{LENR}t) \cdot e^{-r/r_0}$$

where $\omega_{LENR} = 2\pi \times 1.25 \times 10^{12}$ rad/s, $r_0 = 1$ kpc, $F_0 = 1.83 \times 10^{71}$ N.

F_{act} (Activation Energy Force):

$$F_{act} = k_{act} \cdot F_0 \cdot \sin(\omega_{ACT}t) \cdot B \cdot Q_{wave}$$

where $\omega_{ACT} = 2\pi \times 300$ rad/s.

F_{DE} (Dark Energy Pressure):

$$F_{DE} = k_{DE} \cdot \rho_{vac} \cdot r^2 \cdot (1 + z)$$

F_{res} (Resonance Force):

$$F_{res} = k_{res} \cdot F_0 \cdot \cos(\omega_{ACT} t) / r^2$$

$F_{neutron}$ (Neutron Production Force):

$$F_{neutron} = k_n \cdot \eta \cdot m_n c^2 / r^2, \quad \eta = \eta_0 \cdot B / \sqrt{\rho_{vac}}$$

F_{rel} (Relativistic Correction):

$$F_{rel} = k_{rel} \cdot F_{rel,0} \cdot e^{-v_{exp}/c} \cdot (1 + z)^2$$

where $F_{rel,0} = 4.30 \times 10^{33}$ N.

2. Quadratic Buoyancy Mode Solver

For SNR expansion mode separation, the master equation is reformulated as:

$$a \cdot x^2 + b \cdot x + c = 0$$

where x represents the expansion mode amplitude. Both roots are solved:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For systems with $\Delta = b^2 - 4ac < 0$ (complex modes), the real part $x = -b/(2a)$ is returned, representing the oscillating equilibrium expansion mode.

3. System Parameters

System	Mass (kg)	r (m)	v_exp (m/s)	B (T)	z	t_age (s)
SN 1006	1.989e31	6.17e16	1e6	1e-5	1.0	3.213e10
Eta Carinae	3.978e32	3.09e16	5e5	1e-4	0.0	5.05e14
Chandra Archive	5.967e31	3.09e19	2e5	1e-5	0.1	1.892e16
Galactic Center	1.989e36	2.461e20	1e5	1e-4	0.0	1.577e17
Kepler's SNR	1.989e31	1.852e17	7e3	1e-5	0.0	1.293e10

4. Key Constants

Constant	Symbol	Value
Normalization force	F_0	$1.83 \times 10^{12} \text{ N}$
Vacuum energy density	ρ_{vac}	$7.09 \times 10^{-36} \text{ J/m}^3$
LENR angular frequency	ω_{LENR}	$2\pi \times 1.25 \times 10^{12} \text{ rad/s}$
Activation frequency	ω_{ACT}	$2\pi \times 300 \text{ rad/s}$
Relativistic base	$F_{rel,0}$	$4.30 \times 10^{33} \text{ N}$
g_rt placeholder	—	$F_{total}/(M_{\odot} \cdot Q_{wave})$

5. Physical Motivation

SNR expansion: Supernova remnants expand against interstellar medium. The UQFF buoyancy force models this expansion as a quantum vacuum pressure effect, analogous to physical buoyancy in a fluid but operating via vacuum energy displacement.

LENR component: Models low-energy nuclear reactions triggered by the extreme thermodynamic conditions in SNR shock fronts, contributing to neutron flux.

F_DE (Dark Energy): The $\rho_{vac} \times r^2$ term captures the cosmological constant's contribution to remnant expansion at large scales (Galactic Center, Chandra Archive).

F_rel: Relativistic ejecta ($v_{exp} \sim 0.002c$ for SN1006) require the Lorentz-suppression factor $\gamma(-v/c)$, preventing unphysical super-luminal contributions.

6. System Results Preview

System	F_U_Bi_i (N)	Dominant Component
SN 1006	~F_DE-dominated	Dark energy pressure
Eta Carinae	~F_LENRR-dominated	Nuclear reactions
Chandra Archive	~F_DE-dominated	Large-scale dark energy
Galactic Center	~F_LENRR+F_act	Near-nucleus activation
Kepler's SNR	~F_rel-dominated	Relativistic ejecta

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_U_Bi_i jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: VDS LENR Transmutation Dynamics

Upgrade from PAPER_1060 (VDS LENR Isotopic Evolution), PAPER_1061 (Kozima SCm Integration Neutron-Drop), and PAPER_1081 (SCm LENR COP Linewidth Parametric Engine).

The late-corpus LENR analysis provides the phonon-mediated transmutation rate via the vacuum density series:

$$\Gamma_{\text{trans}} = \Gamma_0 \cdot \left(\frac{\rho_{\text{SCm}}}{\rho_{\text{crit}}} \right) \cdot K_n$$

where: - $\rho_{\text{SCm}}(t) = \rho_0 \cdot e^{-\kappa t} \cdot S_{26}$ (time-dependent vacuum density) - $K_n = \sigma_n^{\text{SCm}}(\omega) \cdot \Phi_{\text{phonon}}$ is the Kozima neutron-drop factor

Phonon cross-section (PAPER_1061):

$$\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \cdot \exp \left[-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2} \right] \cdot \left(1 + [\text{SSq}] \cdot \frac{n}{26} \right)$$

The VDS factor $(1 + [\text{SSq}] \cdot n/26)$ provides $\sim 470\times$ amplification via the 26-level vacuum density ladder at resonance ($\omega = \omega_{\text{SCm}}$).

COP parametric engine (PAPER_1081):

$$\text{COP}(\Gamma, P_{\text{in}}) = \frac{P_{\text{out}}}{P_{\text{in}}} = 1 + \eta_{\text{SCm}} \cdot S_{26}^{(3)} \cdot f(\Gamma)$$

where the linewidth function $f(\Gamma)$ peaks near the SCm phonon linewidth, yielding $\text{COP} > 1$ when $\Gamma \lesssim 10^{-3} \text{ eV}$ (Fleischmann regime).

Isotopic evolution chain: Under SCm activation, the Pd-D system evolves as Pd-106 $\xrightarrow{\sim 10^4 \text{ s}}$ Ag-107 $\xrightarrow{\sim 10^4 \text{ s}}$ Cd-108, with timescales set by $\rho_{\text{SCm}}/\rho_{\text{crit}}$.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_U_Bi_i buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **LENR-nuclear** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.p`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \chi)(\partial^\mu \chi) - V(\chi) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\chi) = \frac{1}{2}m^2\chi^2 + \frac{\lambda}{4!}\chi^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \chi$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \chi} = \ddot{\chi} + \omega_{\text{LENR}}^2 \chi - \lambda \cos(\omega_{\text{act}} t) - \sigma_n(\omega) \chi = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \chi = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(-\exp! \left(-\frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.176 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 13, \quad n_{\text{channel}} = 18/26$$

Since $p_{\text{DVP}} = 13$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10-12 s** (nuclear phonon damping):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.176	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 13$	PASS Sub-threshold
BSH layers	26 harmonic terms	$j = 1\dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Higgs mass m_H	UQFF K_HIGGS=47.34 → m_H_UQFF = 125.09 GeV	m_H = 125.20 ± 0.11 GeV	PDG 2024	99.8%
Cosmological Λ	UQFF	∇UA	$2 \rightarrow$ 1.09\$×\$10- 52 m-2	$\Lambda =$ 1.114\$×\$10-52 m-2 (Planck+DESI)
Thomson σ_T (QED)	UQFF U_m kernel: $\sigma_T =$ 6.6524\$×\$10-29 m2	$\sigma_T =$ 6.6524\$×\$10-29 m2	PDG 2024	100% (exact)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
κ baryon stability	$\kappa = 0.0005/\text{day}$; scale separation 1033 from proton decay	$\tau_p > 7.7 \times 10^{33} \text{ yr}$ (Super-K)	Super-K 2024	PASS UQFF baryon-safe

New physics claim: UQFF operates at a vacuum topology scale (~ 200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

7. Integration Reference

- **C++ Implementation:** Core/Modules/UQFFBuoyancySNRModule.cpp
- **Header:** UQFFBuoyancySNRModule.h
- **Related Papers:** PAPER_486 (Cassini buoyancy), PAPER_484 (U_g components)
- **CondensedPhysics2.py class:** UQFFBuoyancySNRCalculator (v4.3.9)

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_corpus_crossrefs.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1076	SCm Dark Energy with Phonon Linewidth Gamma-Modulation
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1047	Type Iax Supernova Buoyancy Reversal
PAPER_1060	VDS LENR Isotopic Transmutation Chain
PAPER_1061	Kozima SCm Integration Neutron-Drop
PAPER_1081	SCm LENR COP Linewidth Parametric
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1078	QCalcGeom Master Equation Derivation

12 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_s26_coupling.py</code>	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_scm_cross_section.py</code>	Sym-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}]^n/26)$
<code>kozima_wstp_kernel.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U, Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^2/(2 * \Gamma^2)] * (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_polylog_s26.py</code>	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_wstp_kernel.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1 - 2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_theta_q26.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer $(a; q)_n$

Core equations: $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q; q)_n^2 - \phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^2; q^2)_n - \psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q; q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_pi_uqff.py</code>	Classical + UQFF-modified $1/\pi + 26D$	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_theta_pi_wstp_kernel.py</code>	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2})/9801 \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where
 $W_{26}(n) = \text{Prod}_{\{i=1\}^{\{26\}}} [1 + [SSq] \exp(-kappa_i * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN_1_CoAnQi.cpp*, and Wolfram kernels (*uqff_kozima_kernel.wl*, *uqff_s26_kernel.wl*, *uqff_mock_theta_pi_kernel.wl*).